

Tackling Real-World Data Integration Challenges through Generic RML Extensions

Els de Vleeschauwer¹[0000–0002–8630–3947]

IDLab, Department of Electronics and Information Systems, Ghent University –
imec, Technologiepark-Zwijnaarde 122, 9052 Ghent, Belgium
`els.de_vleeschauwer@ugent.be`

Abstract. To construct RDF Knowledge Graphs (KGs), heterogeneous source data can be systematically transformed and published in downstream systems. Declarative mapping languages enable generic, transparent, and maintainable KG construction. However, real-world data systems require access control for different levels of granularity, apply data references across and within data sources, and apply deeply-nested and mixed-format data structures. These realities cannot be handled in current mapping languages, which hinders their broader adoption and leads practitioners to rely on custom pipelines. This PhD research aims to improve KG construction to support the aforementioned realities in the RDF Mapping Language (RML), a mature declarative mapping language. Guided by the Design-oriented Information Systems Research methodology, I identify functional gaps in RML (namely, data-driven publication, explicit relations across and within data sources, integration of individual data values from other data sources, integration of mixed-format data, and references to structural positions) while working on data integration projects, specify and implement RML extensions, evaluate their effectiveness and efficiency, and disseminate them for feedback, adoption, and standardization. As preliminary findings I introduced (i) RML Dynamic Targets for data-driven publication in different levels of granularity (e.g., per-entity or per-policy) and (ii) RML Reference Conditions to improve performance when referencing data across data sources (e.g., decreasing external RML engine Morph-KGC's execution time by 12%). I am currently elaborating RML Logical Views, an expressive data access description, integrating data value integration, mixed-format data structures, and structural position references. My involvement in the W3C KG Construction Community allows me to validate the proposed solutions with a broad set of real-world use cases, and the integration of my solutions in an existing and mature specification increases the impact of my work. An important risk in my research is managing its scope, as each new use case may reveal additional gaps. I will focus on the set of gaps identified so far as the minimal viable scope required to meaningfully complete this PhD.

Keywords: data integration · knowledge graph construction · RDF Mapping Language.

Middle Stage Ph.D.

1 Introduction/Motivation

RDF Knowledge Graphs (KGs) provide structured data with explicit, machine-interpretable semantics [36]. To construct KGs, existing heterogeneous source data can be systematically transformed to RDF data and published into downstream data management systems. Declarative mapping languages provide several benefits for KG construction [28]. First, the KG construction and publication process can be described in a generic, reproducible, and semantic manner, independent of any particular implementation (i.e., the *mapping engine*). This avoids vendor lock-in and enables users to select or switch between engines as requirements evolve (e.g., changes in data volume, performance needs, or architectural constraints). Second, declarative mappings provide metadata about the KG (e.g. using links between mapping rules and resulting RDF triples as provenance information [11], and deriving quality indicators [12]). Finally, these languages come with an ecosystem of tools that support not only the execution of mappings but also the maintenance, evolution, debugging, and documentation of the KG production process. Given these benefits, wider adoption of declarative mapping languages can improve the efficiency, transparency, and robustness of KG construction workflows.

A recent survey on declarative mapping languages [28] revealed the RDF Mapping Language (RML) as a mature and widely adopted solution, and as the only solution supporting the end-to-end pipeline from data source to published RDF data. The *initial RML specification*, introduced in 2014 [13] extended the Relational to RDF Mapping Language (R2RML) [5] to generate RDF from not only relational databases but also other types of data sources such as CSV, JSON, and XML. Since then, numerous extensions have been proposed to address real-world requirements [2,6,7,9,10,19,25,29]. The W3C KG Construction Community Group (KGC Community)¹ consolidated the limitations of R2RML and the early RML versions into a set of mapping challenges², and addressed multiple of those mapping challenges in a new *modular RML specification*, released in 2023 [18], incorporating some of the aforementioned extensions [6,9,19,29]. Despite RML’s maturity, some mapping challenges remain unsolved, and new challenges² and (partial) solutions [30] are continuously reported by the KGC Community.

2 Problem Statement and Contributions

Functional gaps in current declarative mapping languages hinder their adoption. Real-world data integration projects require granular permissioned publication, large-scale and flexible joining of data entities, support for mixed-format data representations, and the interpretation of implicit structural information. Such requirements exceed the expressive capabilities of current declarative mapping

¹ <https://www.w3.org/community/kg-construct/>

² <https://github.com/kg-construct/mapping-challenges>

languages, particularly RML. As a result, practitioners resort to ad hoc pre-processing, post-processing, or custom pipelines for (parts of) the KG construction process [26], which undermines the advantages of declarative specifications and leads to opaque and hard-to-reproduce data integration workflows.

This PhD addresses the following research problem: *The modular RML specification lacks expressive power to declaratively model the incoming and outgoing interfaces required in real-world KG construction pipelines.* This results in the following research objective: *Improving the expressiveness of RML, specifically focusing on the interaction with existing real-world data management systems.*

The starting point for my research is the modular RML specification introduced in 2023 [18] and backed by the KGC Community. By focusing on such a mature declarative mapping language, I aim to maximize impact. To scope the work, I focus on the incoming and outgoing interface between the KG construction process and existing systems: how existing heterogeneous data is accessed, combined, and interpreted during KG construction (modelled as a *logical source* in RML), and how constructed RDF is published into downstream data management systems (modelled as a *logical target* in RML).

Below, I formulate specific functional gaps that I will tackle. How I identified these gaps will be further explained in Section 3.

Gap 1 (outgoing): data-driven publication. The usage of Semantic Web technologies for data integration has extended from open data sharing to permissioned data sharing, as exemplified by standardization efforts from, for example, the Linked Web Storage Working group³ that continues the work from the Solid project⁴ and the Fedora Repository⁵. More broadly, this is taken up in the deployment of data spaces. This permissioned data sharing context requires data-driven publication of outputs (e.g., publishing per-entity or per-policy artifacts). RML defines *static* logical targets to describe the interface with downstream data management systems, without the option to dynamically—i.e., based on source data—define the output targets, limiting RML’s applicability in data space-related contexts.

Gap 2 (incoming): explicit relations across and within data sources. When joining data from two tables within one relational database, a significant performance gain can be achieved: certain joins can assume *referential integrity*, i.e., the join defined in the mapping is a join supported by the source database. When you can assume referential integrity within the data source you are mapping, the mapping engine does not need to double-check these references and redundant processing can be skipped. RML does not allow the description of referential integrity or other constraints across and within data sources, limiting the potential performance of the mapping engine needed for large-scale integration pipelines.

³ <https://www.w3.org/groups/wg/lws/>

⁴ <https://solidproject.org/>

⁵ <https://fedorarepository.org/>

Gap 3 (incoming): integration of individual data values from other data sources. RML constructs triples (or quads) with data values from one data source. Enrichment of that data with another data source typically involves adding (i) relations to other concepts (object properties) and (ii) data values to the same concept (data properties). Object properties are expressed as join constructs in RML, however, RML provides no functionality to add data properties using data originating from other data sources. Additionally, data from different data sources cannot be combined to construct one RDF term and, when integrating hierarchical data, the hierarchical relations between values originating from different levels of the hierarchy is not preserved.

Gap 4 (incoming): integration of mixed-format data. Data management systems may manage mixed-format data, e.g., embedding JSON objects in a specific column of relational database table [24] or encoding line vectors as WKT strings in a document store. RML only supports single-format data sources.

Gap 5 (incoming): references to structural positions. The structural position of values within the data (e.g., the first, second, or third row of a CSV file, or the first, second, or third object in a JSON array) may define a data concept's identity or its provenance. RML provides no mechanism to reference such structural positions. This limits the ability to generate identifiers derived from the data's structural context, relate RDF terms back to specific locations in the source for debugging or to create provenance, or coordinate multiple rules that target the same structural position within a source.

For each identified gap, the following research questions are addressed:

RQx.1: What requirements must be met to bridge this gap in RML?

RQx.2: Which minimal language extension satisfies these requirements?

RQx.3: What trade-offs does this extension introduce in terms of language complexity and performance?

My research contributions consist of the *specifications and implementations of RML extensions* developed in response to these questions. These contributions broaden RML's applicability across diverse data integration scenarios and provide input for the standardization efforts of the upcoming RML working group.

3 Research Methodology and Approach

I adopt the Design-oriented Information Systems Research methodology [37], which aims to develop and deliver an artefact as a research contribution that addresses a problem situated in a real-world context. The artefact is created and evaluated through iterative refinement cycles conducted in close collaboration with stakeholders, e.g., companies, organizations, and governmental agencies. The Design-oriented Information Systems Research methodology is guided by four core principles, requiring the artefact to be *abstracted* (applicable to a broader class of problems), *original* (substantially contributing to the advancement of knowledge through publications and accumulated experience), *justified*

(comprehensibly grounded and accommodating validation), and *beneficial* (providing immediate or future value to the relevant stakeholder groups) [8].

Through requirements analysis of real-world use cases within collaborative research & development (R&D) projects, I catalogue and analyse existing KG construction gaps (focusing on interfaces), devise and implement solutions, and evaluate their effectiveness and efficiency to the problem at hand in collaboration with the relevant organizations and the broader KG construction community. The resulting artefacts are specifications and their implementations.

My approach and evaluation plan aligns with the four phases of the Design-oriented Information Systems Research methodology (Analysis, Design, Evaluation, and Dissemination⁶) [37]. **RQx.1** and **RQx.2** will be tackled during the Analysis phase (**P1**) and the Specification Design phase (**P2a**), respectively. The Implementation Design phase (**P2b**) prepares the Evaluation phase (**P3**), which tackles **RQx.3**.

3.1 Phase 1: Analysis (P1)

I frequently perform R&D projects in the scope of data integration, typically in collaboration with industry or government. When constructing a KG for such data integration projects, I encounter (interfacing) limitations of RML to meet the project requirements. During this phase, I analyse the project requirements and related RML limitations. Basing my work on collaborative R&D projects makes sure the produced artefact is *beneficial*. Then, to make sure the produced artefact is *abstract* and *original*, I collect similar issues within the KGC Community (sourced from RML Challenges, issues in the RML-modules, and input from the KGC Community⁷) and (partial) solutions proposed in related work. The gaps as presented at Section 2 are a result of this Analysis phase.

3.2 Phase 2: Design (P2)

Given the scope of my work (declarative mappings languages) I split this phase into two: specification design (P2a) and specification implementation (P2b).

Specification design (P2a) In this phase, I translate the identified gaps into RML extensions that address the functional requirements uncovered during Analysis. As mapping languages such as RML are declarative, their specifications ensure *abstraction*: they describe a conceptual solution that can be instantiated in multiple implementations. To maintain coherence with the existing RML specification, I follow the established modular RML design approach [18]: the specification is accompanied by an OWL ontology defining the conceptual model, SHACL shapes formalizing its constraints, and test cases demonstrating expected behaviour and supporting future conformance testing. Where relevant,

⁶ Renamed from the original coined term 'Diffusion'.

⁷ As can be found, e.g., at <https://github.com/kg-construct/use-cases>, <https://github.com/kg-construct/rml-questions/discussions>, or <https://github.com/search?q=desc&q=org%3Akg-construct&s=created&type=Issues>.

I provide a formal definition of the specifications (e.g., building on existing mapping algebra research [22]) to ensure unambiguous interpretation and provable optimizations, thereby *justifying* the design.

Implementation design (P2b) To operationalize the designed specification, I develop a reference implementation. This implementation serves multiple purposes: it enables systematic evaluation of the specification in realistic settings (*justified*); it ensures that the solution is directly applicable within and beyond the original project context (*beneficial*); and it provides a concrete blueprint that can inspire and guide adoption in existing RML engines (*original*). All implementations are released as open-source software with comprehensive documentation. This guarantees transparency, reproducibility, and accessibility for the KGC Community, while fostering collaborative refinement and long-term sustainability of the artefact.

4 Evaluation Plan

My evaluation plan covers the final two phases of the Design- oriented Information Systems Research methodology: Evaluation and Dissemination. **RQx.1** and **RQx.3** will be evaluated in the Evaluation phase through functional and performance evaluation. **RQx.2** will be evaluated in the Dissemination phase through community feedback and peer review.

4.1 Phase 3: Evaluation

I evaluate the developed artefacts’ effectiveness and, where relevant, efficiency.

Effectiveness (P3a) To assess functional effectiveness, I examine to which extent the implemented specification resolves the identified gaps through test cases. In addition, I develop demonstrators that illustrate the supported functionalities within realistic KG construction scenarios. These demonstrators rely on representative project data and configurations to ensure that the evaluation closely reflects real-world usage. This qualitative assessment verifies whether the specification and its reference implementation meet stakeholder requirements (*beneficial*) and meaningfully extend RML’s expressive power or usability (*original*). It thereby *justifies* the design choices embodied in the extensions.

Efficiency (P3b) When the proposed solutions can be compared to existing approaches, I evaluate their performance in terms of execution time, memory usage, and CPU consumption. This quantitative assessment is carried out using KROWN [27], an established benchmarking framework for reproducible KG construction performance evaluation, and involves established RML engines alongside—or integrated with—the developed reference implementations. Whenever possible, I reuse publicly available benchmark datasets, including the artificial datasets from the KGC Workshop performance tracks⁸ and data stemming from GTFS-Madrid-Bench [4]. When needed, I generate additional arti-

⁸ <https://zenodo.org/records/7689310>, [10.5281/zenodo.11577087](https://zenodo.org/records/11577087)

cial datasets. The resulting measurements quantify the computational trade-offs introduced by the new specification and thus quantify when its use is *beneficial*.

4.2 Phase 4: Dissemination (P4)

First, I start the dissemination by producing peer-reviewed papers. Their acceptance is an important first validation of my work. Second, I integrate the designed specifications into the modular RML specification, through active participation in the KGC Community. Evaluation metrics of this phase (next to publications) are (i) whether the proposed specifications have been incorporated into the modular RML ecosystem, (ii) whether other RML engines have adopted the specification, and (iii) whether the solutions are used in external projects beyond the initial context. These metrics reflect the artefact’s long-term value and its alignment with the *beneficial* and *abstract* principles of the Design-oriented Information Systems Research methodology.

5 Results

As preliminary findings, I introduced (i) RML Dynamic Targets for data-driven publication in different levels of granularity (tackling *Gap 1, outgoing*) and (ii) RML Reference Conditions to remove performance bottlenecks when relating mapping rules enabling RMLStreamer to execute GTFS-Madrid-Bench (tackling *Gap 2, incoming*). I am currently elaborating RML Logical Views, an expressive data access description (jointly tackling *Gaps 2, 3, 4 and 5, incoming*).

5.1 RML Dynamic Targets

I identified Gap 1 while developing a Solid-based data sharing solution for Onto-DESIDE⁹, a Horizon Europe project on data sharing for increased circular economy (**P1**). I specified RML Dynamic Targets to describe the data-driven generation of any publication target¹⁰ and an RML HTTP Request Access description to support HTTP access to any web resource with extensible authentication¹¹ (**P2a**). I implemented these extensions in RMLMapper¹² (**P2b**). I evaluated the effectiveness of the solution through test cases¹³, and validated it on the Onto-DESIDE use cases¹⁴ (**P3a**). My work was published at the KGC Workshop [31],

⁹ <https://ontodeside.eu/>

¹⁰ <https://w3id.org/imec/rml/specs/target/dynamictarget>

¹¹ <https://w3id.org/imec/rml/specs/access/httprequest>

¹² <https://github.com/RMLio/rmlmapper-java/releases/tag/v7.2.0>

¹³ <https://github.com/RMLio/rmlmapper-java/tree/master/src/test/resources/solid-target>

¹⁴ https://ontodeside.eu/wp-content/uploads/2024/10/Onto-DESIDE_WP6_Deliverable_D6.8_final.pdf

can be integrated in the RML-IO registry¹⁵ as a community-driven specification, and is reused in the Solid4Media project¹⁶, a living lab project exploring the value of Solid data vault within a real media context (**P4**).

5.2 RML Reference Conditions

I identified Gap 2 while integrating multiple big datasets for AI4FL¹⁷, an ICON project aiming to improve the end-to-end fresh food supply chain. I proposed an adaptation to the RML specification, introducing a *reference condition* as new RML term (**P2a**). I provided a proof-of-concept implementation, RML-LooseGenerator¹⁸, simulating the effect of reference conditions by rewriting the mapping files (**P2b**). Applying this on data stemming from GTFS-Madrid-Bench [4], I showed that reference conditions do not influence KG generation results when referential integrity is assumed within the source system (**P3a**). Efficiency measurements confirmed that removing the integrity checks leads to faster KG generation. Interpreting the join conditions as reference conditions, RMLStreamer executed the GTFS-Madrid-Bench at scale 100 three times faster than the state-of-the-art RML engine Morph-KGC [34]. Applying reference conditions directly to Morph-KGC decreased its execution time with 12% [34]. RML Reference Conditions was applied to the AI4FL pipeline to transform the provided datasets within the requested time frame [23] (**P3b**). After the publication of this work [32,34], I presented it to the KGC Community. The feedback resulted in integrating the main concept behind RML Reference Conditions as Foreign Key Annotation in a new RML module: RML Logical Views (**P4**).

5.3 RML Logical Views

Whilst processing the feedback on RML Reference Conditions, I initiated a dedicated task force within the KGC Community, guiding biweekly meetings from September 2023 to March 2024. The task force’s initial focus, translated in Gaps 2 and 3, was then expanded, identified as Gaps 4 and 5 (**P1**). The work of this task force resulted in the specification of RML Logical Views as an additional module for RML (**P2a**). I developed a proof-of-concept implementation¹⁹ (**P2b**), which I used to test the feasibility and effectiveness of the specification and to ensure correctness of the test cases (**P3a**). My work was published at the KGC Workshop [33,35]. RML Logical Views is integrated as an additional module in the modular RML specification²⁰ and was included in the KGC Workshop Challenge 2025²¹, leading to implementations in other RML engines [16] (**P4**).

¹⁵ <https://kg-construct.github.io/rml-io-registry/>

¹⁶ <https://www.solid4media.be/>

¹⁷ <https://www.imec-int.com/en/research-portfolio/ai4foodlogistics>

¹⁸ <https://github.com/RMLio/rml-loose-generator>

¹⁹ <https://github.com/RMLio/rml-view-to-csv>

²⁰ <https://w3id.org/rml/lv/spec/>

²¹ <https://doi.org/10.5281/zenodo.14970817>

Meanwhile, I investigate the role of RML Logical Views within the modular RML specification, targeting a publication in the Semantic Web Journal. This future work focuses on formalizing RML Logical Views (**P2a**). There has been substantial recent work on formalizing RML, with some approaches focusing on execution-order optimizations [1,15,17], others on integrating logical and physical planning [14], and others on defining a language-agnostic Mapping Algebra [21,22]. I build on this latter Mapping Algebra because it provides a generic formal foundation that can be extended to ensure unambiguous semantics for the new RML module. I will implement this formalization (**P2b**), validate the formalization by demonstrating its compliance with the RML Logical Views test cases using this implementation (**P3a**), and evaluate the efficiency of this implementation and of RML engines that have integrated (parts of) the RML Logical Views module. This evaluation compares the RML Logical Views approach with alternative solutions in other RML modules and assesses engine performance for features that rely exclusively on this module. The experiments include all RML engines that implemented (a part of) RML Logical Views and at least five state-of-the-art RML engines for the comparison with alternatives solutions in other RML modules (**P3b**). Through this work, I aim to further advance the adoption of RML Logical Views and support its incorporation into the upcoming RML standardization efforts (**P4**).

6 State of the Art

Here, I compare solutions proposed in this PhD with other RML-related proposals that address the identified functional gaps, or, where none exist, with solutions from other declarative mapping languages.

For Gap 1, the state of the art is limited to two end-to-end setups for deploying a Linked Data Platform (LDP) with integrated declarative KGC pipeline. Morph-LDP [20] is an R2RML-based implementation that exposes relational data as RDF data in a predefined LDP structure. LDP-DL [3] is a language to define the design of LDPs, integrating SPARQL-Generate (a SPARQL-based declarative mapping language) to transform heterogeneous data sources to RDF data. Both works wrap a declarative mapping language in an external framework to define the data-driven LPD publication targets. In contrast, RML Dynamic Targets integrates data-driven publication for any heterogeneous target (not limited to LDPs) directly in a declarative mapping language.

For Gap 2, I extend on the main concepts of constraints in Ontop Lenses²².

For Gap 3 and Gap 4, (partial) solutions for RML have been published, but they were not adopted in the modular RML specification. RML Views for Tabular Sources [2] expands RML’s existing SQL views to tabular sources instead of solely relational databases, to support flexible joins and mixed-format data in RML. through SQL queries. RML Logical Views offers an RDF-based solution for heterogeneous sources (not limited to tabular sources). RML Fields [10] proposes

²² <https://ontop-vkg.org/guide/advanced/lenses.html#constraints>

an RDF language construct and algorithm for RML to deal with hierarchical data and mixed-format data in RML. Building on this work, RML Logical Views revises and extends RML Fields, integrating also a solution for flexible joins (to fully cover Gap 3).

For Gap 5, I extend on the notion of indexes, introduced to support the algorithmic representation of RML Fields [10].

7 Conclusions/Lessons Learned

Despite its maturity, RML still needs extensions to satisfy the requirements of real-world use cases. By following the Design-oriented Information Systems Research methodology, I develop specifications and their implementations as concrete artefacts that are abstract, original, justified, and beneficial. RML Dynamic Targets extends RML to support permissioned data sharing and is implemented in RMLMapper. RML Logical Views provides an interface to deeply nested data sources and data sources with mixed formats, allows combining data values from two or more data sources, provides structural references, and allows expressing the relation across and within data sources. RML-view-to-CSV is a proof-of-concept implementation of RML Logical Views, while RMLLooseGenerator simulates the effect of using the foreign key annotations specified in RML Logical Views.

My future work includes a further investigation of RML Logical Views, leading to its formalization and an efficiency evaluation of its implementations, as input for the standardization of RML. Meanwhile, I continue to follow-up on related efforts in this space (e.g., the recent forming of the Façade-X Community Group²³, a SPARQL-based mapping language), to ensure my encountered challenges—and potentially my devised solutions—can be further incorporated in the broader KG construction research domain.

The proposed extensions of RML broadens its applicability, avoiding sub-optimal solutions for KG construction, and thus improving data integration. I learned that, while providing specifications and implementations opens the path for interoperability and reuse of the proposed solutions, real impact requires a close cooperation with a dedicated community, bringing together the ideas of researchers and developers working in the same domain, and, mostly, getting their acceptance and adoption of developed solutions.

An important challenge throughout my PhD is managing its scope: as each new R&D project has the risk of introducing a new challenge, I will focus on the set of gaps identified so far as the minimal viable scope required to meaningfully complete this PhD.

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²³ <https://www.w3.org/community/facade-x/>

References

1. Arenas-Guerrero, J., Chaves-Fraga, D., Toledo, J., Pérez, M.S., Corcho, O.: Morphkgc: Scalable knowledge graph materialization with mapping partitions. *Semantic Web* **15**, 1–20 (2022). <https://doi.org/10.3233/SW-223135>, <https://doi.org/10.3233/SW-223135>
2. Arenas-Guerrero, J., Alobaid, A., Navas-Loro, M., Pérez, M.S., Corcho, O.: Boosting knowledge graph generation from tabular data with RML views. In: *The Semantic Web*, pp. 484–501. Springer Nature Switzerland (2023). https://doi.org/10.1007/978-3-031-33455-9_29
3. Bakerally, N., Zimmermann, A., Boissier, O.: LDP-DL: A Language to Define the Design of Linked Data Platforms. In: Gangemi, A., Navigli, R., Vidal, M.E., Hitzler, P., Troncy, R., Hollink, L., Tordai, A., Alam, M. (eds.) *The Semantic Web*. pp. 33–49. Springer International Publishing, Cham (2018). https://doi.org/10.1007/978-3-319-93417-4_3
4. Chaves-Fraga, D., Priyatna, F., Cimmino, A., Toledo, J., Ruckhaus, E., Corcho, O.: Gtfs-madrid-bench: A benchmark for virtual knowledge graph access in the transport domain. *Journal of Web Semantics* **65**, 100596 (2020). <https://doi.org/10.1016/j.websem.2020.100596>
5. Das, S., Sundara, S., Cyganiak, R.: R2RML: RDB to RDF Mapping Language. Working group recommendation, World Wide Web Consortium (W3C) (Sep 2012), <http://www.w3.org/TR/r2rml/>
6. De Meester, B., Seymoens, T., Dimou, A., Verborgh, R.: Implementation-independent Function Reuse. *Future Generation Computer Systems* **110**, 946–959 (Sep 2020). <https://doi.org/10.1016/j.future.2019.10.006>, <https://doi.org/10.1016/j.future.2019.10.006>
7. Debruyne, C., O’Sullivan, D.: R2rml-f: Towards sharing and executing domain logic in r2rml mappings. In: *Proceedings of the 9th Workshop on Linked Data on the Web*. CEUR Workshop Proceedings, vol. 1593 (2016), <http://ceur-ws.org/Vol-1593/article-13.pdf>
8. Delport, P.M., Von Solms, R., Gerber, M.: Methodological guidelines for design science research. *Procedia Computer Science* **237**, 195–203 (2024). <https://doi.org/https://doi.org/10.1016/j.procs.2024.05.096>, <https://www.sciencedirect.com/science/article/pii/S1877050924011128>, international Conference on Industry Sciences and Computer Science Innovation
9. Delva, T., Arenas-Guerrero, J., Iglesias-Molina, A., Corcho, , Chaves-Fraga, D., Dimou, A.: Rml-star: A declarative mapping language for rdf-star generation. In: *International Semantic Web Conference (ISWC), Posters and Demos*. CEUR Workshop Proceedings, vol. 2980 (2021), <http://ceur-ws.org/Vol-2980/paper374.pdf>
10. Delva, T., Van Assche, D., Heyvaert, P., De Meester, B., Dimou, A.: Integrating nested data into knowledge graphs with RML fields. In: Chaves-Fraga, D., Dimou, A., Heyvaert, P., Priyatna, F., Sequeda, J. (eds.) *Proceedings of the 2nd International Workshop on Knowledge Graph Construction co-located with 18th Extended Semantic Web Conference (ESWC 2021)*. vol. 2873. CEUR (Jun 2021), <http://ceur-ws.org/Vol-2873/paper9.pdf>
11. Dimou, A., De Nies, T., Verborgh, R., Mannens, E., Van de Walle, R.: Automated Metadata Generation for Linked Data Generation and Publishing Workflows. In: Auer, S., Berners-Lee, T., Bizer, C., Heath, T. (eds.) *Proceedings of the Workshop on Linked Data on the Web co-located with 25th International World Wide Web Conference (WWW2016)*. CEUR Workshop Proceedings, vol. 1593.

- CEUR, Montreal, Canada (Apr 2016), http://events.linkedata.org/ldow2016/papers/LDOW2016_paper_04.pdf
12. Dimou, A., Kontokostas, D., Freudenberg, M., Verborgh, R., Lehmann, J., Mannens, E., Hellmann, S., Van de Walle, R.: Assessing and Refining Mappings to RDF to Improve Dataset Quality. pp. 133–149 (2015). https://doi.org/10.1007/978-3-319-25010-6_8, http://link.springer.com/chapter/10.1007/978-3-319-25010-6_8
 13. Dimou, A., Vander Sande, M., Colpaert, P., Verborgh, R., Mannens, E., Van de Walle, R.: RML: A Generic Language for Integrated RDF Mappings of Heterogeneous Data. In: Bizer, C., Heath, T., Auer, S., Berners-Lee, T. (eds.) Proceedings of the 7th Workshop on Linked Data on the Web. CEUR Workshop Proceedings, vol. 1184. CEUR (Apr 2014), http://ceur-ws.org/Vol-1184/ldow2014_paper_01.pdf
 14. Freund, M., Schmid, S., Harth, A.: Efficient knowledge graph construction based on optimized plans. In: Linking Meaning: Semantic Technologies Shaping the Future of AI. Studies on the Semantic Web, vol. 62, pp. 2–18. IOS Press (2025). <https://doi.org/10.3233/SSW250005>
 15. Iglesias, E., Jozashoori, S., Chaves-Fraga, D., Collarana, D., Vidal, M.E.: SDM-RDFizer: An RML Interpreter for the Efficient Creation of RDF Knowledge Graphs. In: Proceedings of the 29th ACM International Conference on Information & Knowledge Management. ACM (Oct 2020). <https://doi.org/10.1145/3340531.3412881>
 16. Iglesias, E., Vidal, M.E.: Results for knowledge graph creation challenge 2025: Sdm-rdfizer. In: Chaves-Fraga, David and Dasoulas, Ioannis and Debruyne, Christophe and Dimou, Anastasia and Serles, Umutcan and Van Assche, Dylan (ed.) KGCW 2025 : Knowledge Graph Construction. vol. 3999, p. 8. CEUR (2025)
 17. Iglesias, E., Jozashoori, S., Vidal, M.E.: Scaling up knowledge graph creation to large and heterogeneous data sources. *Journal of Web Semantics* **75**, 100755 (2023). <https://doi.org/10.1016/j.websem.2022.100755>, <https://doi.org/10.1016/j.websem.2022.100755>
 18. Iglesias-Molina, A., Van Assche, D., Arenas-Guerrero, J., De Meester, B., Debruyne, C., Jozashoori, S., Maria, P., Michel, F., Chaves-Fraga, D., Dimou, A.: The RML Ontology: A Community-Driven Modular Redesign After a Decade of Experience in Mapping Heterogeneous Data to RDF. In: Proceedings of the International Semantic Web Conference (ISWC). pp. 152–175. Lecture Notes in Computer Science, Springer, Cham (2023). https://doi.org/10.1007/978-3-031-47243-5_9
 19. Michel, F., Djimenou, L., Faron-Zucker, C., Montagnat, J.: Translation of relational and non-relational databases into rdf with xr2rml. In: WEBIST 2015 - Proceedings of the 11th International Conference on Web Information Systems and Technologies. pp. 443–454. SciTePress, Lisbon, Portugal (2015). <https://doi.org/10.5220/0005448304430454>
 20. Mihindukulasooriya, N., Priyatna, F., Corcho, O., García-Castro, R., Esteban-Gutiérrez, M.: morph-LDP: An r2rml-based linked data platform implementation. In: Lecture Notes in Computer Science, pp. 418–423. Springer International Publishing (2014). https://doi.org/10.1007/978-3-319-11955-7_59
 21. Min Oo, S., De Meester, B., Taelman, R., Colpaert, P.: Algebraic mapping operators for knowledge graph generation. *SEMANTIC WEB* **16**(5), 29 (2025), <http://doi.org/10.1177/22104968251361350>
 22. Min Oo, S., Hartig, O.: An algebraic foundation for knowledge graph construction. In: Curry, Edward and Acosta, Maribel and Poveda-Villalón, Maria and van Erp,

- Marieke and Ojo, Adegbayegba and Hose, Katja and Shimizu, Cogan and Lisen, Pasquale (ed.) SEMANTIC WEB, ESWC 2025, PT I. vol. 15718, pp. 3–22. Springer Cham (2025), http://doi.org/10.1007/978-3-031-94575-5_1
23. Moura, H.D., de Vleeschauwer, E., Haesendonck, G., De Meester, B., D’eer, L., De Schepper, T., Mercelis, S., Mannens, E.: Performance of an end-to-end inventory demand forecasting pipeline using a federated data ecosystem. In: ENGINEERING PROCEEDINGS. vol. 68, p. 10 (2024), <http://doi.org/10.3390/engproc2024068033>
 24. Petkovic, D.: Comparison of techniques for storing json data in relational form. *International Journal of Computer Applications* **185**, 9–12 (07 2023). <https://doi.org/10.5120/ijca2023922962>
 25. Slepicka, J., Yin, C., Szekely, P., Knoblock, C.A.: Kr2rml: An alternative interpretation of r2rml for heterogeneous sources. In: *Proceedings of the 6th International Workshop on Consuming Linked Data (COLD 2015)*. CEUR Workshop Proceedings, vol. 1426 (2015), <http://ceur-ws.org/Vol-1426/paper-08.pdf>
 26. Tailhardat, L., Chabot, Y., Troncy, R.: Designing noria: a knowledge graph-based platform for anomaly detection and incident management in ict systems. In: *Proceedings of the 4th International Workshop on Knowledge Graph Construction (KGCW 2023) co-located with 20th Extended Semantic Web Conference (ESWC 2023)* (2023), <https://ceur-ws.org/Vol-3471/paper3.pdf>
 27. Van Assche, D., Chaves-Fraga, D., Dimou, A.: KROWN : a benchmark for RDF graph materialisation. In: Demartini, Gianluca and Hose, Katja and Acosta, Mariabel and Palmonari, Matteo and Cheng, Gong and Skaf-Molli, Hala and Ferranti, Nicolas and Hernández, Daniel and Hogan, Aidan (ed.) SEMANTIC WEB-ISWC 2024, PT III. vol. 15233, pp. 20–39. Springer (2025), http://doi.org/10.1007/978-3-031-77847-6_2
 28. Van Assche, D., Delva, T., Haesendonck, G., Heyvaert, P., De Meester, B., Dimou, A.: Declarative RDF graph generation from heterogeneous (semi-)structured data: A systematic literature review. *Journal of Web Semantics* (2022). <https://doi.org/10.1016/j.websem.2022.100753>
 29. Van Assche, D., Haesendonck, G., De Mulder, G., Delva, T., Heyvaert, P., De Meester, B., Dimou, A.: Leveraging Web of Things W3C Recommendations for Knowledge Graphs Generation. In: *Web Engineering, 21st International Conference, ICWE 2021, Proceedings*. pp. 337–352 (2021). https://doi.org/10.1007/978-3-030-74296-6_26, <https://dylanvanassche.be/assets/pdf/icwe2021-wot-logical-target.pdf>
 30. Van Assche, D., Julián, A.R., De Meester, B., Colpaert, P.: Incremental knowledge graph construction from heterogeneous data sources. *Semantic Web Journal* (2024)
 31. de Vleeschauwer, E., Haesendonck, G., De Meester, B., Colpaert, P.: Extending RML to support permissioned data sharing with multiple views. In: Chaves-Fraga, David and Dasoulas, Ioannis and Debruyne, Christophe and Dimou, Anastasia and Serles, Umutcan and Van Assche, Dylan (ed.) KGCW 2025 : Knowledge Graph Construction. vol. 3999, pp. 1–15. CEUR (2025)
 32. de Vleeschauwer, E., Haesendonck, G., Van Assche, D., De Meester, B.: RML-Streamer with Reference Conditions in the KGCW Challenge 2023. In: Chaves-Fraga, D., Dimou, A., Iglesias-Molina, A., Serles, U., Van Assche, D. (eds.) *Proceedings of the 4th International Workshop on Knowledge Graph Construction co-located with 20th Extended Semantic Web Conference ESWC 2023*. vol. 3471 (2023), <https://ceur-ws.org/Vol-3471/paper11.pdf>
 33. de Vleeschauwer, E., Maria, P., De Meester, B., Colpaert, P.: RML-view-to-CSV: A Proof-of-Concept Implementation for RML Logical Views. In: Chaves-Fraga, D.,

- Dimou, A., Iglesias-Molina, A., Serles, U., Van Assche, D. (eds.) Proceedings of the 5th International Workshop on Knowledge Graph Construction co-located with 21th Extended Semantic Web Conference (ESWC 2024). CEUR Workshop Proceedings, vol. 3718. CEUR-WS.org (2024), <https://ceur-ws.org/Vol-3718/paper2.pdf>
34. de Vleeschauwer, E., Min Oo, S., De Meester, B., Colpaert, P.: Reference Conditions: Relating Mapping Rules Without Joining. In: Proceedings of the 4rd International Workshop on Knowledge Graph Construction (KGCW 2023) co-located with 20th Extended Semantic Web Conference (ESWC 2023). p. 8 (May 2023), <https://ceur-ws.org/Vol-3471/paper10.pdf>
 35. de Vleeschauwer, E., Van Assche, D., De Meester, B.: RMLMapper supported by RML-view-to-CSV in the KGCW Challenge 2025. In: Chaves-Fraga, David and Dasoulas, Ioannis and Debruyne, Christophe and Dimou, Anastasia and Serles, Umutcan and Van Assche, Dylan (ed.) KGCW 2025 : Knowledge Graph Construction. vol. 3999, p. 3. CEUR (2025)
 36. Wagner, R., Kitzelmann, E., Boersch, I.: Mitigating hallucination by integrating knowledge graphs into LLM inference - a systematic literature review. In: Zhao, J., Wang, M., Liu, Z. (eds.) Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop). pp. 795–805. Association for Computational Linguistics, Vienna, Austria (Jul 2025). <https://doi.org/10.18653/v1/2025.acl-srw.53>, <https://aclanthology.org/2025.acl-srw.53/>
 37. Österle, H., Becker, J., Frank, U., Hess, T., Karagiannis, D., Krcmar, H., Loos, P., Mertens, P., Oberweis, A., Sinz, E.J.: Memorandum on design-oriented information systems research. *European Journal of Information Systems* **20**(1), 7–10 (2011). <https://doi.org/10.1057/ejis.2010.55>, <https://doi.org/10.1057/ejis.2010.55>